



CS302 Midterm FILE 1 Solved BY TEAM HADI

Digital Logic and Design (Virtual University of Pakistan)

VIRTUAL UNIVERSITY OF PAKISTAN

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TEAM HADI(VU LAHORE)

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WARNING: Team HADI is not responsible for any mistake or wrong answer. All students reading and using this document may check and confirm the answers at their own.



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Best of luck!



Question No : 1 of 26

Marks: 1 (Budgeted Time 1 Min)

$(A+B)(A+C) =$ _____

Answer (Please select your correct option)

☐ B+C

☐ A+BC

correct



☐ AB+C

☐ AC+B

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Question No : 2 of 26

Marks: 1 (Budgeted Time 1 Min)

Excess-8 code assigns _____ to "-8"

Answer (Please select your correct option)

☐ 1110

☐ 1100

☐ 1000

☐ 0000

correct

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Question No : 3 of 26

Marks: 1 (Budgeted Time 1 Min)

The output of the expression $F = A.B.C$ will be Logic _____ when $A=1$, $B=0$, $C=1$.

Answer (Please select your correct option)

☐ Undefined

☐ One

☒ Zero

correct



☐ No Output as input is invalid.

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Question No : 4 of 26

Marks: 1 (Budgeted Time 1 Min)

The cell marked 6 in 4-variable K-Map represent minterm 6 or the maxterm 6 having the following binary value of variables A, B, C and D.

Answer (Please select your correct option)

☐ A=1, B=1, C=0, D=0

☒ A=0, B=1, C=1, D=0

☐ A=0, B=0, C=1, D=1

☐ A=1, B=0, C=0, D=1



correct

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Question No : 5 of 26

Marks: 1 (Budgeted Time 1 Min)

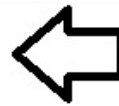
A comparator circuit has multiple inputs and _____ outputs

Answer (Please select your correct option)

☐ 2

☐ 3

correct



☐ 4

☐ 7

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Question No : 6 of 26

Marks: 1 (Budgeted Time 1 Min)

The output $A < B$ is set to 1 when the input combinations is _____

Answer (Please select your correct option)

☐ A=10, B=01

☐ A=11, B=01

☐ A=01, B=01

☐ A=01, B=10

correct



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Question No : 7 of 26

Marks: 1 (Budgeted Time 1 Min)

The _____ Encoder is used as a keypad encoder.

Answer (Please select your correct option)

- ☐ 2-to-8 encoder
- ☐ 4-to-16 encoder
- ☐ BCD-to-Decimal
- ☐ Decimal-to-BCD Priority

correct



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Question No : 8 of 26

Marks: 1 (Budgeted Time 1 Min)

Using multiplexer as parallel to serial converter requires _____ connected to the multiplexer

Answer (Please select your correct option)

☐ A parallel to serial converter circuit

correct



☐ A counter circuit

☐ A BCD to Decimal decoder

☐ A 2-to-8 bit decoder

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Question No : 9 of 26

Marks: 1 (Budgeted Time 1 Min)

The four outputs of two 4-input multiplexers, connected to form a 16-input multiplexer, are connected together through a 4-input _____ gate

Answer (Please select your correct option)

☐ AND

☒ OR

☐ NAND

☐ XOR

correct



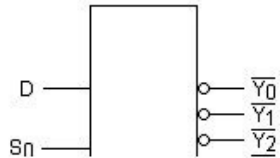
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Question No : 10 of 26

Marks: 1 (Budgeted Time 1 Min)

The device shown here is most likely a _____



Answer (Please select your correct option)

- ☐ Comparator
- ☐ Multiplexer
- ☐ Demultiplexer
- ☐ Parity generator

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Question No : 11 of 26

Marks: 1 (Budgeted Time 1 Min)

GAL is an acronym for _____.

Answer (Please select your correct option)

☐ Generic Analysis Logic

☐ Giant Array Logic

☐ General Array Logic

☐ Generic Array Logic



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Question No : 12 of 26

Marks: 1 (Budgeted Time 1 Min)

In the binary number "10011" the weight of the most significant digit is ____

Answer (Please select your correct option)

☐ 2^4 (2 raise to power 4)

correct

☐ 2^3 (2 raise to power 3)

☐ 2^0 (2 raise to power 0)

☐ 2^1 (2 raise to power 1)

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Question No : 13 of 26

Marks: 1 (Budgeted Time 1 Min)

The output of the expression $F=A+B+C$ will be Logic _____ when $A=0$, $B=1$, $C=1$.

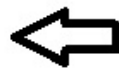
Answer (Please select your correct option)

☐ Undefined

☐ One

☐ Zero

☒ 10 (binary)



correct

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Question No : 14 of 26

Marks: 1 (Budgeted Time 1 Min)

Which of the following simplification method is intended to be used for Boolean expressions having more than four variables?

Answer (Please select your correct option)

☐ Boolean Algebra and rules

☐ Karnaugh Map

☒ Quine-McCluskey



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☐ Demorgan's Theorem

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Question No : 15 of 26

Marks: 1 (Budgeted Time 1 Min)

A particular half-adder has _____ inputs and _____ output(s).

Answer (Please select your correct option)

☐ 3, 1

☐ 3, 2

☐ 2, 1

☐ 2, 2

correct

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Question No : 16 of 26

Marks: 1 (Budgeted Time 1 Min)

A quantity having continuous values is a/an _____ quantity.

Answer (Please select your correct option)

☐ digital

☐ binary

☐ analog

correct



☐ natural

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Question No : 17 of 26

Marks: 1 (Budgeted Time 1 Min)

Which one of the following expressions is an example of Commutative Law for Multiplication?

Answer (Please select your correct option)

☐ $AB+C = A+BC$

☐ $A(B+C) = B(A+C)$

☒ $AB=BA$

correct



☐ $A+B=B+A$

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Question No : 18 of 26

Marks: 1 (Budgeted Time 1 Min)

When four 1's are taken as a group on a Karnaugh map, the number of variables eliminated from the output expression is/are _____

Answer (Please select your correct option)

☐ 1

☐ 2

☐ 3

☐ 4

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Question No : 19 of 26

Marks: 1 (Budgeted Time 1 Min)

Identify the Boolean expression that is in standard POS form.

Answer (Please select your correct option)

☐ $(A + \bar{B})(\bar{A} + B + \bar{C})$

correct



☐ $\bar{A}\bar{B}C + A\bar{B}C$

☐ $(A + \bar{B} + \bar{C})(A + B + C)$

☐ $(A + B)(A + C)$

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Question No : 20 of 26

Marks: 1 (Budgeted Time 1 Min)

Simplifying POS (Product of sum) expression requires the grouping of _____

Answer (Please select your correct option)

☐ 0s
correct ←

☐ 1s

☐ Don't care

☐ Both 0s and 1s

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Question No : 21 of 26

Marks: 2 (Budgeted Time 4 Min)

How many input and output bits do a Half-Adder contain?

Answer ([Please click here to Add Answer](#))



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Question No : 22 of 26

Marks: 2 (Budgeted Time 4 Min)

Draw the Truth-Table of NAND based S-R Latch.

Answer (Please [click here](#) to Add Answer)



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Question No : 23 of 26

Marks: 3 (Budgeted Time 6 Min)

Draw the block diagram of 4-bit BCD Adder.

Answer ([Please click here to Add Answer](#))



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Question No : 24 of 26

Marks: 3 (Budgeted Time 6 Min)

Tri-state buffer control input can be connected in four different ways. Name any three of them.

Answer (Please [click here](#) to Add Answer)



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Question No : 25 of 26

Marks: 5 (Budgeted Time 10 Min)

Simplify the following Boolean expressions, using Boolean Algebraic rules

- $(A+B)(A+B')$
- $(A+B)'(A'+B')'$

Answer ([Please click here to Add Answer](#))



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Marks: 5 (Budgeted Time 10 Min)

Answer (Please [click here](#) to Add Answer)



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